## Power Series, Taylor Series, & Maclaurin Series

If $c_0, c_1, c_2, \ldots$ are constants and $x$ is a variable, then a series of the form

$$\sum_{k=0}^{\infty} c_k x^k = c_0 + c_1 x + c_2 x^2 + \cdots + c_k x^k + \cdots \tag{3}$$

is called a ***power series in x***. Some examples are

$$\sum_{k=0}^{\infty} x^k = 1 + x + x^2 + x^3 + \cdots$$

$$\sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

**9.8.1 DEFINITION** If $f$ has derivatives of all orders at $x_0$, then we call the series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + \cdots \tag{1}$$

the ***Taylor series for f about x = x₀***. In the special case where $x_0 = 0$, this series becomes

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(k)}(0)}{k!}x^k + \cdots \tag{2}$$

in which case we call it the ***Maclaurin series for f***.

Note that the $n$th Maclaurin and Taylor polynomials are the $n$th partial sums for the corresponding Maclaurin and Taylor series.

▶ **Example 2** Find the Taylor series for $1/x$ about $x = 1$.

***Solution.*** In Example 6 of Section 9.7 we found that the $n$th Taylor polynomial for $1/x$ about $x = 1$ is

$$\sum_{k=0}^{n} (-1)^k (x - 1)^k = 1 - (x - 1) + (x - 1)^2 - (x - 1)^3 + \cdots + (-1)^n (x - 1)^n$$

Thus, the Taylor series for $1/x$ about $x = 1$ is

$$\sum_{k=0}^{\infty} (-1)^k (x - 1)^k = 1 - (x - 1) + (x - 1)^2 - (x - 1)^3 + \cdots + (-1)^k (x - 1)^k + \cdots \quad ◀$$

Find the Taylor series of $f(x) = \ln x$ about $x = 1$ up to cubic terms.

**Step 1: Write Taylor formula**

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f^{(3)}(a)}{3!}(x-a)^3 + \cdots$$

Here $a = 1$

**Step 2: Compute derivatives**

$$f(x) = \ln x$$

$$f'(x) = \frac{1}{x}, \quad f''(x) = -\frac{1}{x^2}, \quad f^{(3)}(x) = \frac{2}{x^3}$$

**Step 3: Evaluate at $x = 1$**

$$f(1) = 0, \quad f'(1) = 1, \quad f''(1) = -1, \quad f^{(3)}(1) = 2$$

**Step 4: Substitute**

$$\ln x = (x-1) - \frac{(x-1)^2}{2} + \frac{2}{3!}(x-1)^3 + \cdots$$

$$\boxed{\ln x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} + \cdots}$$

▶ **Example 1** Find the Maclaurin series for

(a) $e^x$ (b) $\sin x$ (c) $\cos x$ (d) $\dfrac{1}{1-x}$

***Solution (a).*** In Example 2 of Section 9.7 we found that the $n$th Maclaurin polynomial for $e^x$ is

$$p_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!}$$

Thus, the Maclaurin series for $e^x$ is

$$\sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^k}{k!} + \cdots$$

***Solution (b).*** In Example 3(a) of Section 9.7 we found that the Maclaurin polynomials for $\sin x$ are given by

$$p_{2k+1}(x) = p_{2k+2}(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + (-1)^k \frac{x^{2k+1}}{(2k+1)!} \quad (k = 0, 1, 2, \ldots)$$

Thus, the Maclaurin series for $\sin x$ is

$$\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + (-1)^k \frac{x^{2k+1}}{(2k+1)!} + \cdots$$

***Solution (c).*** In Example 3(b) of Section 9.7 we found that the Maclaurin polynomials for $\cos x$ are given by

$$p_{2k}(x) = p_{2k+1}(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots + (-1)^k \frac{x^{2k}}{(2k)!} \quad (k = 0, 1, 2, \ldots)$$

Thus, the Maclaurin series for $\cos x$ is

$$\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots + (-1)^k \frac{x^{2k}}{(2k)!} + \cdots$$

***Solution (d).*** In Example 5 of Section 9.7 we found that the $n$th Maclaurin polynomial for $1/(1-x)$ is

$$p_n(x) = \sum_{k=0}^{n} x^k = 1 + x + x^2 + \cdots + x^n \quad (n = 0, 1, 2, \ldots)$$

Thus, the Maclaurin series for $1/(1-x)$ is

$$\sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \cdots + x^k + \cdots \blacktriangleleft$$

**Find the Maclaurin series of $f(x) = \sin x$ up to $x^5$.**

**Step 1: Write Maclaurin formula**

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \cdots$$

**Step 2: Derivatives**

$$f(x) = \sin x$$

$$f'(x) = \cos x, \quad f''(x) = -\sin x, \quad f^{(3)}(x) = -\cos x, \quad f^{(4)}(x) = \sin x$$

**Step 3: Evaluate at 0**

$$f(0) = 0, \quad f'(0) = 1, \quad f''(0) = 0, \quad f^{(3)}(0) = -1, \quad f^{(5)}(0) = 1$$

**Step 4: Substitute**

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\boxed{\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots}$$

Practice Questions

1. Find the Taylor series of $e^x$ about $x = 1$ up to $(x-1)^3$.
2. Find the Taylor series of $\sqrt{x}$ about $x = 1$ up to quadratic terms.
3. Expand $\ln x$ about $x = 2$ up to cubic terms.
4. Find the Taylor series of $x^3$ about $x = 1$ up to $(x-1)^3$.
5. Find the Taylor polynomial of degree 2 for $\cos x$ about $x = \pi/2$.

6. Find the Maclaurin series of $e^x$ up to $x^4$.
7. Find the Maclaurin series of $\cos x$ up to $x^4$.
8. Find the Maclaurin series of $\ln(1+x)$ up to $x^3$.
9. Find the Maclaurin series of $\tan x$ up to $x^3$.
10. Find the Maclaurin series of $\sqrt{1+x}$ up to $x^3$.

11. Find the Maclaurin series of $e^{2x}$ up to $x^3$.
12. Find the Maclaurin series of $\sin(x^2)$ up to $x^6$.
13. Find the Taylor series of $\ln x$ about $x = 1$ up to $(x-1)^4$.
14. Expand $\frac{1}{1-x}$ as a Maclaurin series up to $x^5$.
15. Find the Taylor series of $\sqrt{x}$ about $x = 4$ up to linear terms.